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Informal Ex. 21

Upper bound in ordered set U $\to \mathbb{N}$
maximum in subset S

$U = \mathbb{N}$
$S = \{1, 3, 5, 7\}$

$10 \in U$ upperbound of S, but not in S bc $10 \notin S$.
$7 \in S$ upperbound of S, but also a maximum bc $7 \in S$.

$\mathbb{N} = \mathbb{Z}$
$Q = [-1, 0)$

By Lemma 26, 0 cannot be a maximum b/c it is in $\mathbb{N}$ but not in Q.

Ex 19

Thm $\to$ Min is unique. {WTS $m = m'$}

Pf: Let S be a set w/minima m and m'.

We see m & $m' \in S$ by def. (of minimum)

Further, also by def $m \leq x \ \forall x \in S$.

In particular, $m \leq m'$. $\to$ $m < m'$ or $m = m'$

On the other hand, $m' \leq x \ \forall x \in S$, so $m' \leq m$. $\to$ $m' < m$ or $m' = m$

By trichotomy $m = m'$. $\square$

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Thm 29

Every upperbounded nonempty subset of the naturals has a max.

Pf Sketch:

Let T be such a set, & write B for an upper bound.

Thus if $x \in T$ then $x \leq B$.
Likewise $x \geq 0$ since $T \subset \mathbb{N}$.

So $T \subset \{0 \ldots B\} = \{n \in \mathbb{N} : 0 \leq n \leq B\}$

Thm 28

T has a maximum.

DEF 11: An ordered set U is well ordered if every non-empty subset has a min.

$\mathbb{N}$

Thm 30

"Well-Ordering Principle"

The natural numbers are well ordered.

remark → the well ordering principle is equivalent to induction (Ax 4)

Pf:

Let $S \subseteq \mathbb{N}$ be a non-empty subset.

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$\boxed{\text{Thm 30}}$ "Well-Ordering Principle"

The natural numbers are well ordered.

remark → the well ordering principle is equivalent to induction (Ax 4)

Pf:

Let $S \subseteq \mathbb{N}$ be a non-empty subset.

Let L be the set of lower bounds of S.

note → $0 \in L$ since $0 \leq x \ \forall \ x \in \mathbb{N}$ & $S \in \mathbb{N}$.

In particular, $L \neq \emptyset$.

Since $S \neq \emptyset$ there exists $k \in S$.

If $l \in L$, then $l \leq k$ since each element of L is a lower bound.

We see k is an upper bound for L. Thm. 29 says L has a max; call it m. greatest lower bound

We wts $m \in S$.

For a $\Rightarrow\Leftarrow$, suppose $m \notin S$. Thus $m < k$ for every $k \in S$.

By thm 14, $m+1 \leq k$ or $m+1 \in k$ for every $k \in S$.

$m < k < m+1$

Thus $m+1 \in L$ and $m < m+1$ contradicting the maximality of m. $\cancel{\Rightarrow\Leftarrow}$ $m \in S$. $\square$